Homework 8: Networks (15 points)

Due to Gradescope by 11:45 PM on Tuesday, November 1st
You need to submit a pdf to Gradescope; failure to assign questions to pages will result in a 10% deduction on your grade

Homework Goal: Understand how the Internet works including the five network layers, various protocols, and analyzing network diagrams

1. Practice your vocab from this chapter! (1 point)
   (a) ISP stands for Internet Service Provider, of which there are 3 (number) tiers.
   (b) The ABQ algorithm is the basis for all Data Link Control protocols that are currently used. This algorithm utilizes a(n) ACK that is sent by the receiver to indicate that they received a packet correctly.
   (c) A bridge has knowledge about the nodes connected to it while a repeater just amplifies and forwards the signal.

2. List the five network layers and the goal of each one. (1 point)

Physical - Create a bit pip between two computers
Data Link Layer - Create an error free bit pipe
Network Layer - deliver a message from source to destination
Transport Layer - Create a program-to-program delivery service
Application Layer - Provide rules to offer end-user services

3. Name the two sub-layers of Layer 2. (0.5 points)

2a: Medium Access Control
2b: Logical Link Control

4. What are TCP and UDP, and what is the key difference between them? Give an example of where you would use each. (1.5 points)

TCP is slower and more reliable it would be used when sending an email

UDP is faster however less reliable it would be used in FPS games where you need fast connections and can miss a few frames

5. Why do we need ARQ at both the data link layer and the transport layer? (1 point)

We need ARQ at both the data link layer and the transport layer because at the data link layer it helps to ensure node to node transmission. At the transport layer when using TCP it ensures program to program transmission.

6. Why does the Medium Access Control Protocol require computers to wait a *random* amount of time before attempting retransmission after a collision occurs? (0.5 points)

The MAC Protocol requires computers to wait a certain amount of time as to avoid collision a second time because if two things are trying to retransmit then one will have a shorter wait time.

7. You found a cool GIF online that you want to send to your family members. It's a 4-second color (RGB) GIF with 24 frames per second. Each frame is 1920 x 1080 pixels. The GIF format supports 1 byte per pixel. (4 points)

(a) How many total bits are you going to transmit? Leave your answer in bits.

$1920 \cdot 1080 \cdot 8 = 16,588,800$

$16,588,800 \cdot 24 \cdot 4 = 1,592,524,800$

(b) Based on your answer to part (a), how long will it take to transmit the video to your brother on his 56 Kibps modem (dial up)? Leave your answer in *minutes*.

$$\frac{1,592,524,800}{56 \cdot 2^{10}} = \frac{27771.4}{60} = 463 \text{ min}$$

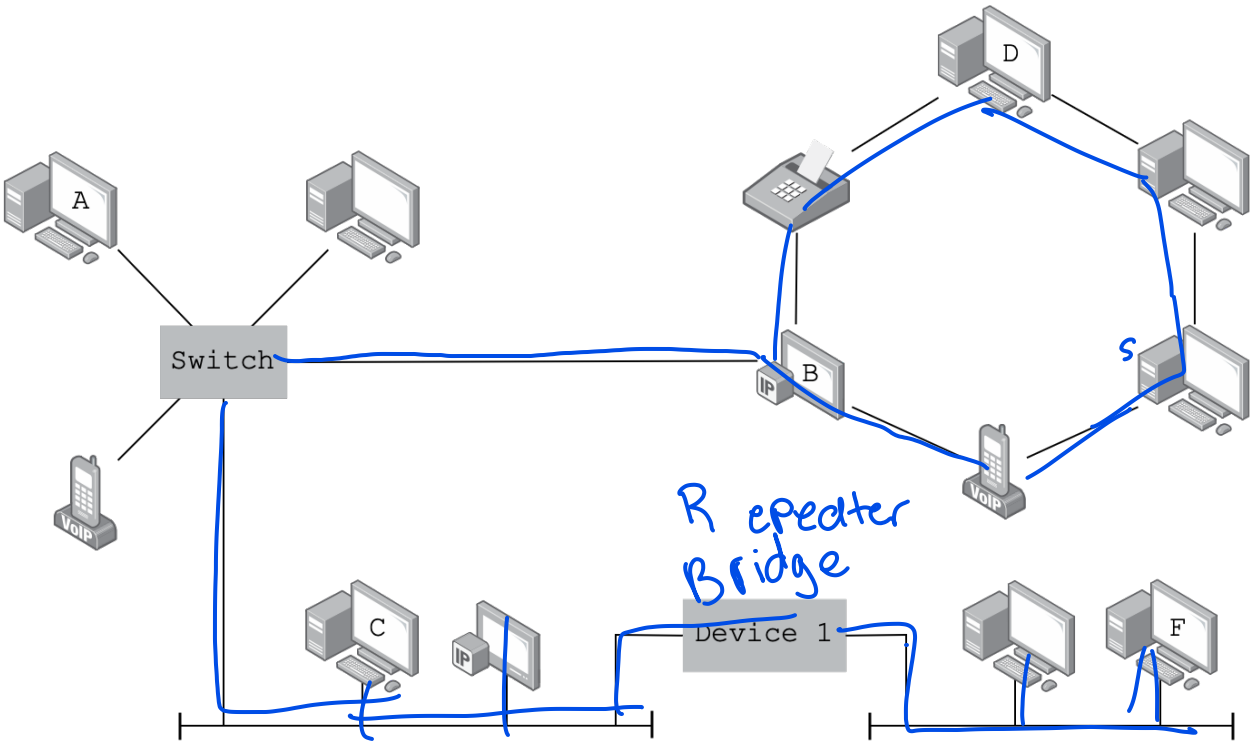
(c) What about sending to your sister on her 2 Mibps DSL line (an older DSL line)? Leave your answer in *minutes*.

$$\frac{1,592,524,800}{2 \cdot 2^{20}} = \frac{759.375}{60} = 13 \text{ min}$$

(d) What about sending to your parents on their 10 Gibps Ethernet link (10-gigabit Ethernet)? Leave your answer in *seconds*.

$$\frac{1,592,524,800}{10 \cdot 2^{30}} = 0.15 \text{ sec}$$

8. Use the network diagram for the following questions. (4 points)
When counting nodes, do not include the sender or any devices (switches, bridges, repeaters).



(a) Identify the 3 different network topologies that are shown in the figure.

Bus, Ring, Star

(b) Suppose A wants to send a message to D. How many nodes could see the message?

8

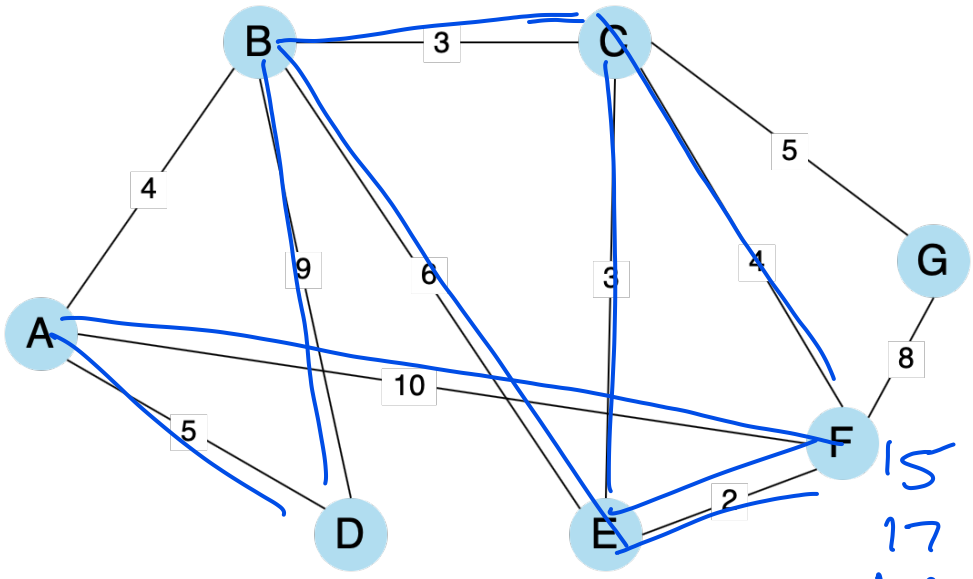
(c) Suppose Device 1 is a bridge and A wants to send a message to C. How many nodes could see the message?

10

(d) Suppose Device 1 is a repeater and C wants to send a message to D. How many nodes could see the message?

9

9. Given the following network, the numbers represent the delay for each link. (1.5 points)



15
17
16
17

(a) List all non-looping paths that take exactly 4 hops from A to G. *Recall, BCGF is 3 hops as it takes 3 edges to get from B to F along that path.*

ABCFG
ABEFG
ADBCG
ABECG

(b) What is the *fastest* (not necessarily *shortest*) path from D to F? What is the total delay of this path?

ADF; 15

(c) Suppose node A goes down. Does this change the fastest path from D to F? If so, what is the new fastest path and what is its total delay?

Yes because now you have to use another not

DBCF; 16